Supplementary Notes: Math Foundations and Extended Explanations

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Contents

1 Introduction **2**

2 Threshold Model **2**
2.1 Exchangeability . . . 2
2.2 Copula . . . 2
2.2.1 Definition . . . 2
2.2.2 Examples of Copulas . . . 4
2.3 Brownian Motion . . . 5
2.4 Geometric Brownian Motion . . . 5
2.5 Itô's Lemma . . . 6

3 Mixture Model **7**
3.1 Random Vectors and Their Distributions . . . 7
3.2 Moments . . . 9

4 Loss Distribution **12**
4.1 Coherent Risk Measures . . . 12
4.2 Basel II Capital Requirement Framework . . . 13
4.2.1 Credit Risk . . . 14
4.2.2 Market Risk . . . 14
4.2.3 Operational Risk . . . 15

5 CreditMetrics and CreditRisk+ **15**
5.1 The Poisson Distribution . . . 15
5.2 Gamma Distribution . . . 16
5.3 Compound Poisson-Gamma Distribution . . . 17
5.4 Default Correlation in CreditRisk+ Framework . . . 19

1 Introduction

2 Threshold Model

2.1 Exchangeability

Exchangeability is a fundamental concept in probability and statistics, particularly in Bayesian inference and the study of random processes. It describes a property of a sequence of random variables where the joint probability distribution remains unchanged under any permutation of the sequence. It is a powerful concept in probability and statistics, enabling the modeling of complex dependencies while maintaining symmetry and simplifying analysis. Exchangeability is a weaker condition than independence, allowing for symmetric dependence among random variables. In other words, the order of the variables does not affect their joint distribution.

Definition 2.1: Exchangeability

A sequence of random variables $X_1, X_2, \ldots, X_n$ is said to be exchangeable if for any permutation π of the indices $\{1, 2, \ldots, n\}$, the joint probability distribution satisfies:

$$(X_1, X_2, \ldots, X_n) \stackrel{d}{=} (X_{\pi(1)}, X_{\pi(2)}, \ldots, X_{\pi(n)})$$

Here, $\stackrel{d}{=}$ means "equal in distribution." This means that the joint distribution of the variables is invariant under reordering.

If the random variables are independent and identically distributed (i.i.d.), they are exchangeable. However, exchangeability does not imply independence. Exchangeable variables can exhibit dependence, but the dependence must be symmetric. Consider a sequence of random variables $X_1,\ X_2,\ X_3$ representing the outcomes of three coin flips. If the sequence is exchangeable, then: $P(X_1 = H, X_2 = T, X_3 = H) = P(X_1 = H, X_2 = H, X_3 = T) = P(X_1 = T, X_2 = H, X_3 = H)$, and so on. The probability depends only on the number of heads and tails, not on their order.

2.2 Copula

2.2.1 Definition

A copula is a multivariate probability distribution for which the marginal probability distribution of each variable is uniform on the interval $[0, 1]$. The key idea is that a copula "couples" the marginal distributions of random variables to form their joint distribution. To understand how it works, we start from *Probability Transformation*. Let X be a random variable with cumulative distribution function F, where F is continuous. Then:

$$U = F(X) \sim U(0, 1)$$

This means that applying F to X transforms it into a uniform $[0, 1]$ random variable. This is because $F(X)$ represents the probability that X takes a value less than or equal to itself, which is always between 0 and 1. Now, we look at *Quantile Transformation*. Suppose $U \sim U(0, 1)$, and let F^{-1} (also called the quantile function or inverse CDF) be

the function that returns the value of X corresponding to a given probability U. Then:

$$X = F^{-1}(U) \sim F$$

This means we can generate a random variable with any desired distribution F by applying its inverse cumulative distribution function to a uniform $[0, 1]$ variable. Consider two random variables X_1 and X_2, with cumulative distribution functions F_1 and F_2, respectively. Using the probability transformation:

$$U_1 = F_1(X_1) \sim U(0, 1), \quad U_2 = F_2(X_2) \sim U(0, 1)$$

This means that the original variables X_1 and X_2 can be mapped to the uniform space $[0, 1]$. The copula function C then captures how U_1 and U_2 are related:

$$C(u_1, u_2) = P(U_1 \leq u_1, U_2 \leq u_2)$$

Since the relationship between X_1 and X_2 is preserved through their ranks (probabilities), the copula completely describes their dependence structure, regardless of their marginal distributions. The rigorous definition is shown below.

Definition 2.2: Copula

A d-dimensional copula is a distribution function on $[0, 1]^d$ with standard uniform marginal distributions. We use the notation $C(\boldsymbol{u}) = C(u_1, \ldots, u_d)$ for copulas. That is to say, C is a mapping of form C: $[0, 1]^d \to [0, 1]$, a mapping of the unit hypercube into the unit interval. The following properties must hold.

(1) $C(u_1, \ldots, u_d) = 0$ if $u_i = 0$ for any i.
(2) $C(1, \ldots, 1, u_i, 1, \ldots, 1) = u_i$ for any $i \in \{1, \ldots, d\}$, $u_i \in [0, 1]$.
(3) $(a_1, \ldots, a_d), (b_1, \ldots, b_d) \in [0, 1]^d$ with $a_i \leq b_i$ we have

$$\sum_{i_1=1}^{2} \cdots \sum_{i_d=1}^{2} (-1)^{i_1+\cdots+i_d} C(u_{1i_1}, \ldots, u_{di_d}) \geqslant 0 \tag{1}$$

where $u_{j1} = a_j$ and $u_{j2} = b_j$ for all $j \in 1, \ldots, d$.

Definition 2.3: Sklar's Theorem

For a joint cumulative distribution function F of random variables $X_1, X_2, \ldots, X_n$ with marginal cumulative distribution functions $F_1, F_2, \ldots, F_n$, there exists a copula C such that:

$$F(x_1, x_2, \ldots, x_n) = C(F_1(x_1), F_2(x_2), \ldots, F_n(x_n)). \tag{2}$$

Sklar's Theorem states that any multivariate joint distribution can be expressed in terms of its marginal distributions and a copula. Conversely, if C is a copula and $F_1, F_2, \ldots, F_n$ are marginal cumulative distribution functions, then F defined above is a valid joint cumulative distribution function. We can show it as follows, for a d-dimensional random vector $(X_1, \ldots, X_d)$ with joint cumulative distribution function $F(x_1, \ldots, x_d)$ and

marginal cumulative distribution functions $F_1, \ldots, F_d$, we can express their joint distribution using a copula:

$$\begin{aligned} F(x_1, \ldots, x_d) &= P(X_1 \leqslant x_1, \ldots, X_d \leqslant x_d) \\ &= P\left(F_1^{-1}(U_1) \leqslant x_1, \ldots, F_d^{-1}(U_d) \leqslant x_d\right) \\ &= P(U_1 \leqslant F_1(x_1), \ldots, U_d \leqslant F_d(x_d)) \\ &= C(F_1(x_1), \ldots, F_d(x_d)) \end{aligned} \tag{3}$$

2.2.2 Examples of Copulas

Copulas can be broadly classified into two categories: implicit and explicit. Implicit copulas arise naturally from multivariate distributions. They are derived from well-known probability distributions such as the normal (Gaussian) distribution and the t-distribution. Explicit copulas, on the other hand, are defined using specific functional forms designed to capture particular features of dependence, such as strong tail dependence in one direction. Examples of explicit copulas include the Clayton copula, which emphasizes dependence in the lower tail (extreme losses), and the Gumbel copula, which focuses on upper tail dependence (extreme gains).

Gaussian Copula. The Gaussian copula is derived from the multivariate normal distribution. Suppose we have a set of standard normal random variables $Z_1, Z_2, \ldots, Z_n$ that follow a joint normal distribution with correlation matrix Σ, such that:

$$(Z_1, Z_2, \ldots, Z_n) \sim \mathcal{N}(0, \Sigma)$$

The Gaussian copula is then given by:

$$C_{\text{Gaussian}}(u_1, u_2, \ldots, u_n) = \Phi_\Sigma(\Phi^{-1}(u_1), \Phi^{-1}(u_2), \ldots, \Phi^{-1}(u_n))$$

where Φ^{-1} is the inverse cumulative distribution function (quantile function) of the standard normal distribution. Φ_Σ is the multivariate normal cumulative distribution function with correlation matrix Σ. The Gaussian copula captures linear correlation but it underestimates joint extreme events (e.g., financial crises).

t-Copula. The t-copula is derived from the multivariate t-distribution and is often used to address the shortcomings of the Gaussian copula. If $T_1, T_2, \ldots, T_n$ follow a multivariate t-distribution with correlation matrix Σ and degrees of freedom ν, the t-copula is given by:

$$C_{\text{t}}(u_1, u_2, \ldots, u_n) = T_{\Sigma,\nu}(T_\nu^{-1}(u_1), T_\nu^{-1}(u_2), \ldots, T_\nu^{-1}(u_n))$$

where T_ν^{-1} is the inverse CDF of the univariate t-distribution with degrees of freedom ν. $T_{\Sigma,\nu}$ is the multivariate t-distribution cumulative distribution function with correlation matrix Σ and ν degrees of freedom. The key advantage of the t-copula is that it exhibits tail dependence, meaning it better captures joint extreme losses in financial markets.

Clayton Copula and Gumbel Copula The Clayton copula is an explicit copula that exhibits strong lower tail dependence, making it ideal for modeling credit risk, where defaults tend to cluster. The Clayton copula is defined as:

$$C_{\text{Clayton}}(u_1, u_2) = \left(\max\left(u_1^{-\theta} + u_2^{-\theta} - 1, 0\right)\right)^{-1/\theta}$$

for $\theta > 0$. The Gumbel copula is another explicit copula but focuses on upper-tail dependence, making it useful for modeling financial markets where asset prices may rise together. It is defined as following for $\theta \geq 1$:

$$C_{\text{Gumbel}}(u_1, u_2) = \exp\left(-\left((-\log u_1)^\theta + (-\log u_2)^\theta\right)^{1/\theta}\right)$$

2.3 Brownian Motion

Brownian motion is a mathematical model that describes the random, continuous movement of particles suspended in a fluid (liquid or gas). It is a foundational concept in probability theory and stochastic processes and is widely used in physics, finance, and other scientific fields.

Key Characteristics of Brownian Motion:

- **Continuous and Random:** The motion is continuous in time, meaning there are no sudden jumps. The movement is random, with no predictable pattern.
- Independent Increments: The movement over non-overlapping time intervals is independent of each other.
- **Normal Distribution of Increments:** Over a small time interval Δt, the change in position ΔW_t follows a normal distribution with:

  $$\Delta W_t \sim \mathcal{N}(0, \Delta t),$$

  where the mean is 0, and the variance is proportional to Δt.
- **Mean and Variance:** The expected position at time t is $\mathbb{E}[W_t] = 0$. - The variance of the position is $\text{Var}(W_t) = t$, showing that the particle's displacement becomes more uncertain over time.
- **Markov Property:** The future state of the process depends only on the current state, not the history.
- **Path Behavior:** The paths of Brownian motion are almost surely continuous but nowhere differentiable.

Definition 2.4: Brownian Motion

Brownian motion W_t is a stochastic process that satisfies: 1. $W_0 = 0$ (the motion starts at zero). 2. W_t has independent increments. 3. For any $t > s$, the increment $W_t - W_s \sim \mathcal{N}(0, t - s)$. 4. W_t has continuous sample paths.

2.4 Geometric Brownian Motion

Geometric Brownian Motion is a widely used stochastic process in finance and economics, particularly for modeling asset prices like stock prices. It describes the evolution of an asset's price over time with both a deterministic trend (drift) and a random fluctuation (volatility).

Definition 2.5: Geometric Brownian Motion

A Geometric Brownian Motion is defined by the following stochastic differential equation (SDE):

$$dV_t = \mu_V V_t \, dt + \sigma_V V_t \, dW_t, \tag{4}$$

where V_t is the asset price at time t. μ_V is the drift (the average rate of return or growth rate of the asset). σ_V is the volatility (the measure of the randomness or uncertainty of the asset's price). W_t is a Brownian motion or Wiener process at time t, which represents the random fluctuations.

To solve the SDE above, we rewrite it using Itô calculus. The key idea is to express the solution V_t in an explicit form. We start with:

$$\frac{dV_t}{V_t} = \mu_V \, dt + \sigma_V \, dW_t.$$

Then integrate both sides from $t = 0$ to $t = T$. The left-hand side involves an integral of $\frac{1}{V_t} dV_t$, which corresponds to the logarithm of V_t. Using Itô's Lemma, the solution becomes:

$$\ln V_T = \ln V_0 + \left(\mu_V - \frac{1}{2}\sigma_V^2\right) T + \sigma_V W_T, \tag{5}$$

where $\frac{1}{2}\sigma_V^2 T$ appears due to the Itô correction term. And then exponentiating both sides gives:

$$V_T = V_0 \exp\left(\left(\mu_V - \frac{1}{2}\sigma_V^2\right) T + \sigma_V W_T\right). \tag{6}$$

The logarithm of V_T is stated in equation (5), where W_T is a Brownian motion at time T, so $W_T \sim \mathcal{N}(0, T)$, meaning it has a mean of 0 and variance T. Multiplying W_T by σ_V scales its variance to $\sigma_V^2 T$. Thus, $\ln V_T$ is normally distributed with: mean of $\ln V_0 + \left(\mu_V - \frac{1}{2}\sigma_V^2\right) T$, and variance of $\sigma_V^2 T$. And the terminal asset value V_T is lognormally distributed because the logarithm of V_T is normally distributed.

2.5 Itô's Lemma

Itô's Lemma is a fundamental result in stochastic calculus that allows us to compute the differential of a function of a stochastic process. It is a key tool in the analysis of stochastic processes, such as Brownian motion or Geometric Brownian Motion.

Definition 2.6: Itô's Lemma

Let X_t be a stochastic process, and let $f(X_t, t)$ be a function that depends on X_t and time t. The differential of $f(X_t, t)$, denoted by $df(X_t, t)$, can be expressed as:

$$df(X_t, t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial X_t} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial X_t^2} (dX_t)^2.$$

Example

Example 2.1 (Special Case: Brownian Motion)**.** *Consider $X_t = W_t$, where W_t is a standard Brownian motion. We know that dW_t is normally distributed with mean zero and variance dt, and $(dW_t)^2 = dt$. In this case, Itô's Lemma becomes:*

$$df(W_t) = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial W_t}dW_t + \frac{1}{2}\frac{\partial^2 f}{\partial W_t^2}dt.$$

Example

Example 2.2 (Special Case: Geometric Brownian Motion)**.** *Let's apply Itô's Lemma to a simple function of a stochastic process, say $f(V_t) = \ln V_t$, where V_t follows a Geometric Brownian Motion. We use Itô's Lemma to compute $d(\ln V_t)$. From the Geometric Brownian Motion equation:*

$$dV_t = \mu_V V_t\, dt + \sigma_V V_t\, dW_t,$$

Itô's Lemma tells us that:

$$d(\ln V_t) = \frac{1}{V_t}dV_t - \frac{1}{2}\frac{1}{V_t^2}(dV_t)^2.$$

Substitute the expressions for dV_t and $(dV_t)^2$, and we get:

$$d(\ln V_t) = \left(\mu_V - \frac{1}{2}\sigma_V^2\right)dt + \sigma_V dW_t.$$

This shows how Itô's Lemma helps simplify the stochastic process by turning a non-linear process (involving V_t) into a linear one in $\ln V_t$, with a drift term and a noise term.

3 Mixture Model

3.1 Random Vectors and Their Distributions

Let $\mathbf{X} = (X_1, \ldots, X_d) : \Omega \to \mathbb{R}^d$ be a d-dimensional random vector (representing, e.g., risk-factor changes). The **(joint) distribution function** F of $\mathbf{X}$ is given by:

$$F(\mathbf{X}) = F_{\mathbf{X}}(\mathbf{x}) = P(\mathbf{X} \leq \mathbf{x}) = P(X_1 \leq x_1, \ldots, X_d \leq x_d), \quad x \in \mathbb{R}^d. \tag{7}$$

The jth **margin** F_j of F or the jth **marginal distribution function** F_j of $\mathbf{X}$ is:

$$\begin{aligned} F_j(x_j) &= P(X_j \leq x_j) = P(X_1 \leq \infty, \ldots, X_{j-1} \leq \infty, X_j \leq x_j, X_{j+1} \leq \infty, \ldots, X_d \leq \infty) \\ &= F(\infty, \ldots, \infty, x_j, \infty, \ldots, \infty), \quad x_j \in \mathbb{R}, \quad j \in \{1, \ldots, d\}. \end{aligned} \tag{8}$$

Similarly for k**-dimensional margins.** Suppose we partition X into (X_1, X_2), where

$$\mathbf{X_1} = (X_1, \ldots, X_k), \quad \mathbf{X_2} = (X_{k+1}, \ldots, X_d),$$

then the **marginal distribution function** of $\mathbf{X_1}$ is given by:

$$F_{\mathbf{X_1}}(\mathbf{x_1}) = P(\mathbf{X_1} \leq \mathbf{x_1}) = F(x_1, \ldots, x_k, \infty, \ldots, \infty).$$

The function F is said to be **absolutely continuous** if:

$$F(x) = \int_{-\infty}^{x_d} \cdots \int_{-\infty}^{x_1} f(z_1, \ldots, z_d)\, dz_1 \ldots dz_d = \int_{(-\infty, \mathbf{x}]} f(\mathbf{z})\, d\mathbf{z}$$

for some density function $f \geq 0$, known as the (joint) density of $\mathbf{X}$ (or F).

A multivariate model for risks $\mathbf{X}$ in the form of a joint distribution function, survival function, or density implicitly describes the dependence of $X_1, \ldots, X_d$. We can then make statements about conditional probabilities. As before, consider $\mathbf{X} = (\mathbf{X_1}, \mathbf{X_2}) \sim F$. The **conditional distribution function** of $\mathbf{X_2}$ given $\mathbf{X_1} = \mathbf{x_1}$ is:

$$F_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{x_1}) = P(\mathbf{X_2} \leq \mathbf{x_2} \mid \mathbf{X_1} = \mathbf{x_1}) = \mathbb{E}\big[I_{\{\mathbf{X_2} \leq \mathbf{x_2}\}} \mid \mathbf{X_1} = \mathbf{x_1}\big], \tag{9}$$

where $\mathbb{E}(\cdot \mid \cdot)$ denotes conditional expectation. A useful identity for conditional distribution functions is:

$$F_{\mathbf{X_1}, \mathbf{X_2}}(\mathbf{x_1}, \mathbf{x_2}) = \int_{(-\infty, \mathbf{x_1}]} F_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{z})\, dF_{\mathbf{X_1}}(\mathbf{z}). \tag{10}$$

If $\mathbf{x_1} \to \infty$, then the marginal distribution of $\mathbf{X_2}$ is given by:

$$F_{\mathbf{X_2}}(\mathbf{x_2}) = \int_{\mathbb{R}^k} F_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{z})\, dF_{\mathbf{X_1}}(\mathbf{z}).$$

If F has a density function f, then the marginal density of $\mathbf{x_2}$ is:

$$f_{\mathbf{X_2}}(\mathbf{x_2}) = \int_{\mathbb{R}^k} f_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{z}) f_{X_1}(\mathbf{z})\, d\mathbf{z}.$$

If F has a density f and $f_{\mathbf{X_1}}$ denotes the density of $\mathbf{X_1}$, then:

$$f(\mathbf{x_1}, \mathbf{x_2}) = \frac{\partial^2}{\partial \mathbf{x_2} \partial \mathbf{x_1}} F(\mathbf{x_1}, \mathbf{x_2}).$$

From this, we obtain:

$$\frac{\partial}{\partial \mathbf{x_2}} F_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{x_1}) f_{\mathbf{X_1}}(\mathbf{x_1}) = f_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{x_1}) f_{\mathbf{X_1}}(\mathbf{x_1}).$$

We define the **conditional density** of $\mathbf{X_2}$ given $\mathbf{X_1} = \mathbf{x_1}$ as:

$$f_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{x_1}) = \frac{f(\mathbf{x_1}, \mathbf{x_2})}{f_{\mathbf{X_1}}(\mathbf{x_1})}. \tag{11}$$

In this case, the **conditional distribution function** $F_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{x_1})$ is given by:

$$F_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{x_1}) = \int_{-\infty}^{\mathbf{x_k}+1} \cdots \int_{-\infty}^{\mathbf{x_d}} f_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{z_{k+1}}, \ldots, \mathbf{z_d} \mid \mathbf{x_1})\, d\mathbf{z_{k+1}} \ldots d\mathbf{z_d}.$$

The random variables $\mathbf{X_1}$ and $\mathbf{X_2}$ are **independent** if: $F(\mathbf{x_1}, \mathbf{x_2}) = F_{\mathbf{X_1}}(\mathbf{x_1}) F_{\mathbf{X_2}}(\mathbf{x_2})$ for all $\mathbf{x_1}, \mathbf{x_2}$.
If F has a density f, then $\mathbf{X_1}$ and $\mathbf{X_2}$ are independent if: $f(\mathbf{x_1}, \mathbf{x_2}) = f_{\mathbf{X_1}}(\mathbf{x_1}) f_{\mathbf{X_2}}(\mathbf{x_2})$ for all $\mathbf{x_1}, \mathbf{x_2}$.
In this case, the conditional density simplifies to: $f_{\mathbf{X_2}|\mathbf{X_1}}(\mathbf{x_2} \mid \mathbf{x_1}) = f_{\mathbf{X_2}}(\mathbf{x_2})$.

3.2 Moments

The **moments** of a random variable X describe various characteristics of its distribution, such as its central tendency and spread. The n-th moment of a random variable X about the origin (also called the raw moment) is defined as:

$$\mu_n = E(X^n) \quad \text{for} \quad n = 1, 2, 3, \ldots \tag{12}$$

For example, the **first moment** $\mu_1 = E(X)$ is the expected value (mean) of X, which represents the central tendency of the distribution. The **second moment** $\mu_2 = E(X^2)$ helps determine the variance when combined with the first moment: $\text{Var}(X) = E(X^2) - (E(X))^2$. The third moment $\mu_3 = E(X^3)$ relates to the skewness of the distribution (whether the distribution is symmetric or skewed to one side). The fourth moment $\mu_4 = E(X^4)$ relates to the kurtosis (the "tailedness" of the distribution). If the moments of X exist, the moment generating function can be used to express them. The **moment generating function** $M_X(t)$ of a random variable X is defined as the expected value of the exponential function of X:

$$M_X(t) = E(e^{tX}) \tag{13}$$

The moment generating function, if it exists, generates the moments of X. The n-th moment can be found by differentiating the MGF and evaluating at $t = 0$:

$$\mu_n = E(X^n) = \frac{d^n}{dt^n} M_X(t)\bigg|_{t=0}$$

The **characteristic function** $\varphi_X(t)$ of a random variable X is a complex-valued function that is similar to the moment generating function but uses the complex exponential instead of the real exponential:

$$\varphi_X(t) = E(e^{itX}) \tag{14}$$

where i is the imaginary unit. The characteristic function is important because it always exists, even for distributions without finite moments. Additionally, it uniquely determines the distribution of X. The characteristic function also provides an alternative way to compute moments. The probability distribution of X can be recovered from its characteristic function using the inversion formula (under certain conditions). And if two random variables X and Y are independent, then their joint characteristic function is the product of their individual characteristic functions. Specifically, the n-th moment of X can be derived from the characteristic function by differentiating with respect to t and evaluating at $t = 0$:

$$\mu_n = E(X^n) = \frac{d^n}{dt^n} \varphi_X(t)\bigg|_{t=0}$$

Example

Example 3.1 (Moments: Beta Distribution)**.** *Calculate the first moment of a Beta distribution.*

Hint: Beta Function Recurrence Property: The Beta function, $\beta(a,b)$, satisfies the following recurrence relation:

$$\beta(a+1,b) = \frac{a}{a+b}\beta(a,b).$$

The probability density function of the Beta distribution with parameters a and b is:

$$g(q) = \frac{1}{\beta(a,b)} q^{a-1}(1-q)^{b-1}, \quad a,b>0, \quad 0<q<1,$$

where $\beta(a,b)$ is the Beta function, defined as:

$$\beta(a,b) = \int_0^1 q^{a-1}(1-q)^{b-1}\,dq.$$

The first moment (mean) of the Beta-distributed random variable $Q \sim Beta(a,b)$ is:

$$\mathbb{E}[Q] = \int_0^1 q \cdot g(q)\,dq.$$

Substitute the expression for $g(q)$:

$$\mathbb{E}[Q] = \int_0^1 q \cdot \frac{1}{\beta(a,b)} q^{a-1}(1-q)^{b-1}\,dq = \frac{1}{\beta(a,b)} \int_0^1 q^{a}(1-q)^{b-1}\,dq.$$

The integral is now of the form of a Beta function. We can thus express the expectation as:

$$\mathbb{E}[Q] = \frac{\beta(a+1,b)}{\beta(a,b)}.$$

Using the property of the Beta function:

$$\frac{\beta(a+1,b)}{\beta(a,b)} = \frac{a}{a+b}.$$

Thus, the mean is:

$$\mathbb{E}[Q] = \frac{a}{a+b}.$$

Example

Example 3.2 (Moments: Beta Distribution)**.** *Calculate the second moment and the variance of a Beta distribution.*

To calculate the second moment $\mathbb{E}[Q^2]$*, we use the following expression:*

$$\mathbb{E}[Q^2] = \int_0^1 q^2 \cdot g(q)\, dq.$$

Substitute the expression for $g(q)$*:*

$$\mathbb{E}[Q^2] = \int_0^1 q^2 \cdot \frac{1}{\beta(a,b)} q^{a-1}(1-q)^{b-1}\, dq = \frac{1}{\beta(a,b)} \int_0^1 q^{a+1}(1-q)^{b-1}\, dq.$$

This is again an integral of the Beta function type:

$$\mathbb{E}[Q^2] = \frac{\beta(a+2,b)}{\beta(a,b)}.$$

Using the property of the Beta function:

$$\frac{\beta(a+2,b)}{\beta(a,b)} = \frac{a(a+1)}{(a+b)(a+b+1)}.$$

Thus, the second moment is:

$$\mathbb{E}[Q^2] = \frac{a(a+1)}{(a+b)(a+b+1)}.$$

The variance $Var(Q)$ *is given by:*

$$Var(Q) = \mathbb{E}[Q^2] - (\mathbb{E}[Q])^2.$$

Substitute the expressions for $\mathbb{E}[Q^2]$ *and* $\mathbb{E}[Q]$*:*

$$Var(Q) = \frac{a(a+1)}{(a+b)(a+b+1)} - \left(\frac{a}{a+b}\right)^2 = \frac{a(a+1)(a+b) - a^2(a+b+1)}{(a+b)^2(a+b+1)}$$

Thus, the variance is:

$$Var(Q) = \frac{ab}{(a+b)^2(a+b+1)}.$$

Additionally, we have

$$\rho_Y = \frac{\mathrm{Var}(Q)}{\mathrm{Var}(Y_i)}.$$

$$\rho_Y = \frac{1}{a+b+1}.$$

4 Loss Distribution

4.1 Coherent Risk Measures

Coherent risk measures are a class of risk measures that satisfy a set of desirable properties, making them suitable for quantifying and managing risk in a consistent and mathematically sound way. A risk measure ρ is said to be **coherent** if it satisfies the following four axioms:

- **Monotonicity.** If a portfolio X always has worse outcomes than portfolio Y, then the risk measure ρ should assign a higher risk to X than to Y. Mathematically, we can write
$$\text{If } X \leq Y \text{ (in all scenarios), then } \rho(X) \geq \rho(Y).$$

- **Subadditivity.** The risk of a combined portfolio $X + Y$ should not exceed the sum of the risks of the individual portfolios X and Y. This suggest diversification reduces risk, so combining portfolios should not increase the overall risk measure. Mathematically, we have
$$\rho(X + Y) \leq \rho(X) + \rho(Y).$$

- **Positive Homogeneity.** Scaling a portfolio by a positive factor λ should scale the risk measure by the same factor. Mathematically, we write
$$\rho(\lambda X) = \lambda \rho(X) \quad \text{for } \lambda \geq 0.$$

- **Translation Invariance.** Adding a risk-free amount c to a portfolio X should reduce the risk measure by c. Mathematically, we have
$$\rho(X + c) = \rho(X) - c.$$

Example

Example 4.1 (VaR is Not Coherent)**.** *Show that VaR fails the subadditivity axiom. Consider two independent portfolios, X and Y, with identical loss distributions. Each portfolio incurs no loss with a 95% probability and a loss of 100 with a 5% probability.*

For both X and Y, the 95% VaR is the smallest loss l such that $P(X > l) \leq 5\%$. From the distributions, $P(X > 0) = 5\%$, so:

$$VaR_{0.95}(X) = VaR_{0.95}(Y) = 0.$$

The combined portfolio $X + Y$ has the following loss distribution:
Loss of 0 *with probability* $0.95 \times 0.95 = 0.9025$
Loss of 100 *with probability* $0.95 \times 0.05 + 0.05 \times 0.95 = 0.095$
Loss of 200 *with probability* $0.05 \times 0.05 = 0.0025$

That is to say, $P(X+Y > 0) = 1-0.9025 = 0.0975$ and $P(X+Y > 100) = 0.0025$. Therefore,

$$VaR_{0.95}(X + Y) = 100.$$

$$VaR_{0.95}(X) + VaR_{0.95}(Y) = 0 + 0 = 0.$$

Since $100 > 0$, we have:

$$VaR_{0.95}(X + Y) > VaR_{0.95}(X) + VaR_{0.95}(Y).$$

This violates the subadditivity axiom:

$$\rho(X + Y) \leq \rho(X) + \rho(Y).$$

4.2 Basel II Capital Requirement Framework

Basel II, introduced by the Basel Committee on Banking Supervision (BCBS) in 2004, refines the capital adequacy framework for banks. It builds upon Basel I by incorporating risk-sensitive approaches to measure capital requirements, ensuring that banks hold adequate capital to cover their risk exposures. The Basel II framework consists of three pillars:

- **Pillar 1:** Minimum Capital Requirements
- **Pillar 2:** Supervisory Review Process
- **Pillar 3:** Market Discipline

Now we focuse on Pillar 1, which establishes the mathematical foundation for determining capital requirements based on credit, market, and operational risks. Under Basel II, banks must maintain a minimum capital adequacy ratio (CAR) of 8% of their risk-weighted assets (RWA). Mathematically, this is expressed as:

$$\frac{\text{Tier 1 Capital} + \text{Tier 2 Capital}}{\text{Risk-Weighted Assets}} \geq 8\%$$

where Tier 1 Capital includes core capital such as common equity and retained earnings. Tier 2 Capital includes supplementary capital such as subordinated debt and hybrid instruments. Risk-Weighted Assets (RWA) represent assets adjusted for risk using predefined weights.

4.2.1 Credit Risk

Credit risk is the potential for loss due to borrower default. Basel II provides three approaches for calculating credit risk capital:

- Standardized Approach. Banks use risk weights assigned by external credit rating agencies (e.g., Moody's, S&P). The capital requirement for a given exposure E is:

$$\text{Capital Requirement} = 8\% \times \sum (E_i \times w_i)$$

  where w_i is the risk weight associated with exposure E_i.

- **Foundation Internal Ratings-Based (F-IRB) Approach.** Under F-IRB, banks estimate the probability of default (PD), while loss given default (LGD) and exposure at default (EAD) are standardized. The capital charge is given by:

$$K = \text{LGD} \times \Phi\left(\frac{\Phi^{-1}(PD) + \sqrt{\rho} \cdot \Phi^{-1}(0.999)}{\sqrt{1-\rho}}\right) \times EAD - PD \times LGD \times EAD$$

  where Φ is the cumulative distribution function of the standard normal distribution. ρ is the asset correlation, defined as:

$$\rho = 0.12 \times \frac{1 - e^{-50PD}}{1 - e^{-50}} + 0.24 \times \left(1 - \frac{1 - e^{-50PD}}{1 - e^{-50}}\right)$$

- Advanced Internal Ratings-Based (A-IRB) Approach. Banks estimate PD, LGD, and EAD internally, allowing for more risk-sensitive capital calculations.

4.2.2 Market Risk

Market risk is the risk of losses due to changes in market variables such as interest rates and exchange rates. Basel II allows banks to calculate market risk capital using:

- Standardized Approach: Uses predefined risk weights for different asset classes.

- Internal Models Approach (IMA): Uses Value at Risk (VaR) models to estimate the potential loss at a confidence level of 99% over a 10-day holding period:

$$\text{Capital Requirement} = \max\left(\text{VaR}_{t-1}, \lambda \times \text{Avg VaR}_{60}\right)$$

  where $\lambda = 3$ is a regulatory multiplier.

4.2.3 Operational Risk

Operational risk covers losses from inadequate internal processes, people, or systems. Basel II offers three methods for calculating operational risk capital:

- Basic Indicator Approach (BIA) as following where $\alpha = 15\%$ (fixed regulatory factor).
$$\text{Capital Requirement} = \alpha \times \text{Gross Income}$$
- Standardized Approach (SA): Banks classify activities into business lines, applying different percentages β_i to each:
$$\text{Capital Requirement} = \sum_i \beta_i \times \text{Gross Income}_i$$
Advanced Measurement Approach (AMA): Uses internal models based on historical loss data.

5 CreditMetrics and CreditRisk+

5.1 The Poisson Distribution

The Poisson distribution is a powerful tool for modeling the number of rare, independent events occurring at a constant average rate. Its simplicity, single-parameter structure, and additivity make it particularly suitable for applications like **CreditRisk+**, where defaults are rare and independent. However, its assumptions (e.g., constant rate, independence) may limit its applicability in more complex scenarios. The Poisson distribution models the probability of a given number of events occurring in a fixed interval of time or space under the following conditions:

- Events occur **independently** of each other.
- The **average rate** λ of occurrence is constant.
- The probability of more than one event occurring in an infinitesimally small interval is negligible.

Definition 5.1: Poisson Distribution

Mathematically, the probability mass function of the Poisson distribution is given by:
$$P(X = k) = \frac{e^{-\lambda}\lambda^k}{k!} \tag{15}$$
Where:

- X: The random variable representing the number of events.
- k: The number of events (a non-negative integer: $k = 0, 1, 2, \ldots$).
- λ: The average rate of occurrence (mean and variance of the distribution).
- e: Euler's number (≈ 2.71828).
- $k!$: The factorial of k.

A key property of the Poisson distribution is that its mean and variance are equal. Specifically, if a random variable follows a Poisson distribution with parameter λ, then both the mean (μ) and variance (σ^2) are given by λ. This equality of mean and variance is a distinctive characteristic of the Poisson distribution, setting it apart from many other probability distributions. Additionally, If $X_1 \sim \text{Poisson}(\lambda_1)$ and $X_2 \sim \text{Poisson}(\lambda_2)$ are independent, then: $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$. This additivity property allows the model to aggregate defaults across different sectors or regions.

Example

Example 5.1 (The number of cars passing through a toll booth)**.** *If the average rate is 10 cars per minute ($\lambda = 10$), the probability of exactly 12 cars passing is:*

$$P(X = 12) = \frac{e^{-10} \cdot 10^{12}}{12!} \approx 0.0948.$$

5.2 Gamma Distribution

The Gamma distribution is a continuous probability distribution that is widely used in statistics, probability theory, and various applied fields. The Gamma distribution is used to model the time until an event occurs (e.g., waiting times) or the sum of exponentially distributed random variables. It is a generalization of the exponential distribution and the chi-squared distribution. The distribution is defined for positive real numbers and is characterized by two parameters:

- Shape parameter (k or α): Controls the shape of the distribution.
- Scale parameter (θ): Controls the spread of the distribution.

Definition 5.2: Gamma Distribution

The probability density function of the Gamma distribution is given by:

$$f(x; k, \theta) = \frac{x^{k-1} e^{-x/\theta}}{\theta^k \Gamma(k)}, \quad x > 0,$$

where:

- x: The random variable (positive real number).
- k: Shape parameter ($k > 0$).
- θ: Scale parameter ($\theta > 0$).
- $\Gamma(k)$: The Gamma function, defined as:

$$\Gamma(k) = \int_0^\infty t^{k-1} e^{-t} \, dt.$$

  For integer values of k, $\Gamma(k) = (k-1)!$.

There are some key properties for Gamma distribution.

- **Support:** The Gamma distribution is defined for $x > 0$.
- **Shape and Scale:** the shape parameter k determines the skewness and mode of the distribution. For $k < 1$: The distribution is highly skewed, with a peak at $x = 0$. For $k = 1$: The Gamma distribution reduces to the exponential distribution. For $k > 1$: The distribution becomes more symmetric and bell-shaped as k increases. The scale parameter θ stretches or compresses the distribution.
- **Mean and Variance:** $\mu = k\theta$, $\sigma^2 = k\theta^2$.
- **Additivity:** if $X_1 \sim \text{Gamma}(k_1, \theta)$ and $X_2 \sim \text{Gamma}(k_2, \theta)$ are independent, then:
$$X_1 + X_2 \sim \text{Gamma}(k_1 + k_2, \theta).$$

Example

Example 5.2 (CreditRisk+)**.** *In CreditRisk+, the latent risk factors Ψ_j are assumed to follow Gamma distributions:*

$$\Psi_j \sim Gamma(\alpha_j, \beta_j), \quad j = 1, \ldots, p.$$

where α_j is the shape parameter. And $\beta_j$is the scale parameter. The parameters are set as:

$$\alpha_j = \sigma_j^{-2}, \quad \beta_j = \sigma_j^2$$

where σ_j^2 represents the variance of the risk factor Ψ_j. The mean of a Gamma-distributed random variable Ψ_j is:

$$E(\Psi_j) = \alpha_j \beta_j = 1.$$

This means that each latent risk factor Ψ_j has an expected value of 1. The variance of a Gamma-distributed random variable Ψ_j is:

$$Var(\Psi_j) = \alpha_j \beta_j^2.$$

$$Var(\Psi_j) = \sigma_j^{-2}(\sigma_j^2)^2 = \sigma_j^2.$$

This means that the variance of Ψ_j is σ_j^2, which is a key parameter in the model.

5.3 Compound Poisson-Gamma Distribution

In this section, we go through the mathematical derivation in detail to show how the unconditional distribution of defaults in **CreditRisk+** becomes a **negative binomial distribution** when the default intensity is **gamma-distributed**. This is a classic example of a compound Poisson-gamma distribution. In CreditRisk+, the number of defaults Y for a portfolio is modeled as a Poisson distribution conditional on the latent factor Ψ(we simplify the setting here by assuming there is only **one latent factor**):

$$Y \mid \Psi = \psi \sim \text{Poi}(\lambda(\psi)),$$

that is,

$$P(Y = y \mid \Psi = \psi) = \frac{e^{-\lambda(\psi)}\lambda(\psi)^y}{y!},$$

where $\lambda(\psi)$ is the default intensity, which depends on the latent factor Ψ. For **a homogeneous portfolio**, the default intensity is:

$$\lambda(\psi) = n \cdot k \cdot \psi,$$

where n is the number of obligors, k is a scaling factor, ψ is the realization of the latent factor Ψ. The latent factor Ψ is modeled as a **gamma-distributed random variable** with mean 1 and variance σ^2:

$$\Psi \sim \text{Gamma}(\alpha, \beta),$$

where $\alpha = \frac{1}{\sigma^2}$ (shape parameter), $\beta = \sigma^2$ (scale parameter). The probability density function of the gamma-distributed latent factor Ψ is:

$$f(\psi) = \frac{\psi^{\alpha-1} e^{-\psi/\beta}}{\beta^\alpha \Gamma(\alpha)},$$

where $\Gamma(\alpha)$ is the gamma function. To find the unconditional distribution of Y, we marginalize over Ψ by integrating the joint distribution of Y and Ψ:

$$P(Y = y) = \int_0^\infty P(Y = y \mid \Psi = \psi) f(\psi)\, d\psi.$$

Substituting the expressions for $P(Y = y \mid \Psi = \psi)$ and $f(\psi)$, we get:

$$P(Y = y) = \int_0^\infty \frac{e^{-nk\psi}(nk\psi)^y}{y!} \cdot \frac{\psi^{\alpha-1} e^{-\psi/\beta}}{\beta^\alpha \Gamma(\alpha)}\, d\psi.$$

Rewrite the integrand, we have

$$P(Y = y) = \frac{(nk)^y}{y! \beta^\alpha \Gamma(\alpha)} \int_0^\infty \psi^{y+\alpha-1} e^{-\psi(nk+1/\beta)}\, d\psi.$$

The integral in the expression is of a form of a **gamma integral**, and its value is:

$$\int_0^\infty \psi^{y+\alpha-1} e^{-\psi(nk+1/\beta)}\, d\psi = \frac{\Gamma(y+\alpha)}{(nk+1/\beta)^{y+\alpha}}.$$

Substituting the gamma integral back into the expression for $P(Y = y)$, we get:

$$P(Y = y) = \frac{(nk)^y}{y! \beta^\alpha \Gamma(\alpha)} \cdot \frac{\Gamma(y+\alpha)}{(nk+1/\beta)^{y+\alpha}}.$$

Simplify the expression:

$$P(Y = y) = \frac{\Gamma(y+\alpha)}{y! \Gamma(\alpha)} \cdot \frac{(nk)^y}{(nk\beta+1)^\alpha (nk+1/\beta)^y}.$$

Let:

$$r = \alpha = \frac{1}{\sigma^2}, \quad p = \frac{1}{1+nk\beta}.$$

Substitute these into the expression:

$$P(Y = y) = \frac{\Gamma(y+r)}{y! \Gamma(r)} \cdot p^r (1-p)^y.$$

This is the probability mass function of the negative binomial distribution:

$$P(Y = y) = \binom{y+r-1}{y} p^r (1-p)^y.$$

5.4 Default Correlation in CreditRisk+ Framework

In this document, we derive the default correlation between two identical obligors within the CreditRisk+ framework. We consider a portfolio of credit obligations where the default intensity for obligor i, denoted as λ_i, is given by:

$$\lambda_i(\psi) = ((1-w) \cdot 1 + w \cdot \psi) \cdot \eta_i \cdot p$$

where:

- w is the weight of the second systematic factor ψ,
- ψ is the second systematic factor, gamma-distributed with mean 1 and standard deviation σ,
- η_i is the idiosyncratic factor, exponentially distributed with parameter 1,
- p is the unconditional default probability (the same for all obligors in the portfolio).

The systematic factor Y is given by:

$$Y = (1-w) \cdot 1 + w \cdot \psi$$

The mean and variance of Y are:

$$\mathbb{E}[Y] = (1-w) + w \cdot 1 = 1$$

$$\mathrm{Var}(Y) = w^2\sigma^2$$

The unconditional default probability p_i for obligor i is:

$$p_i = \mathbb{E}[\lambda_i] = \mathbb{E}[Y \cdot \eta_i \cdot p] = p$$

The covariance between the default indicators D_i and D_j for two obligors i and j is given by:

$$\mathrm{Cov}(D_i, D_j) = \mathbb{E}[D_i D_j] - \mathbb{E}[D_i]\mathbb{E}[D_j]$$

Since D_i and D_j are conditionally independent given Y, we can express the expectation as:

$$\mathbb{E}[D_i D_j] = \mathbb{E}[\mathbb{E}[D_i D_j \mid Y]] = \mathbb{E}[p_i(Y)p_j(Y)]$$

Using the approximation $p_i(Y) \approx Y \cdot \eta_i \cdot p$, we have:

$$\mathbb{E}[D_i D_j] \approx \mathbb{E}[Y^2 \cdot \eta_i\eta_j \cdot p^2] = \mathbb{E}[Y^2] \cdot \mathbb{E}[\eta_i] \cdot \mathbb{E}[\eta_j] \cdot p^2$$

Since $\mathbb{E}[\eta_i] = 1$ and $\mathbb{E}[Y^2] = w^2\sigma^2 + 1$, we get:

$$\mathbb{E}[D_i D_j] \approx (w^2\sigma^2 + 1) \cdot p^2$$

Thus, the covariance is:

$$\mathrm{Cov}(D_i, D_j) \approx (w^2\sigma^2 + 1) \cdot p^2 - p \cdot p = w^2\sigma^2 p^2$$

The variance of the default indicator D_i is:

$$\mathrm{Var}(D_i) = p_i(1-p_i) = p(1-p)$$

Finally, the default correlation ρ_{default} is given by:

$$\rho_{\text{default}} = \frac{\text{Cov}(D_i, D_j)}{\sqrt{\text{Var}(D_i)\text{Var}(D_j)}}$$

Since $\text{Var}(D_i) = \text{Var}(D_j) = p(1-p)$, we have:

$$\rho_{\text{default}} = \frac{w^2\sigma^2p^2}{p(1-p)} = \frac{w^2\sigma^2p}{1-p}$$

The default correlation ρ_{default} represents the correlation between the default events of two identical obligors in the CreditRisk+ framework. This result provides a quantitative measure of the dependence between defaults in the CreditRisk+ framework under the given specifications. It can be used to compare with other credit risk models, such as CreditMetrics, by mapping the default correlation to the asset correlation in those frameworks.